TUTORIAL
Topic: PDA Design (TOC)

PDA Problems

Instruction: Draw both an empty stack PDA and a final state PDA for each of the following languages.

1. **Language:**
$$L = \{a^i b^j c^k \mid i = j \text{ or } j = k\}$$
Instruction: Simulate the string `aabbc`.

2. **Language:**
$$L = \{a^n b^m c^n \mid n \geq 1, m \geq 1\}$$
Instruction: Simulate the string `aabbcc`.

3. **Language:**
Palindromes where the middle symbol must be a
Instruction: Simulate the string `baaab`.

4. **Language:**
Strings over $\{a, b\}$ with no two consecutive a's
Instruction: Simulate the string `ababab`.

5. **Language:**
$$L = \{a^i b^j c^k \mid i + j = k\}$$
Instruction: Simulate the string `aabccc`.

6. **Language:**
$$L = \{a^i b^j c^k \mid i = j \text{ and } j \neq k\}$$
Instruction: Simulate the string `aabbccc`.

7. **Language:**
Strings where:
   - a's and b's alternate
   - start and end with the same symbol

   Instruction: Simulate the string `ababa`.

8. **Language:**
$$L = \{a^n b^m \mid n = m + 2\}$$
Instruction: Simulate the string `aaab`.